Environment Similarity Index: Quantifying map-environment divergences for robot localization performance estimation and improvement

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Abstract— **Should take about this much space: abc.**

I. INTRODUCTION

Robust, lifelong localization is a fundamental ability for all Autonomous Mobile Robots (AMRs). For many years, the gold standard to achieve this has been to use some variation of Monte Carlo Localization, as it strikes a good balance between cost and performance. It can perform global localization, and is multi-modal. For indoor industrial applications, it works well in many cases. It does however, have a major drawback: It assumes the environment is static. As soon as it' surroundings have changed too much, the particle filter diverges and fails, because it does not update it's representation. Making a system that is robust to change is still an ongoing challenge (ref). Some would argue that Simultaneous Localization and Mapping (SLAM) is the best way to solve this; but SLAM suffers in reliability. No system is perfect, and sooner or later, an error will find its way into the system. The resulting erroneous pose will propagate into the resulting map, reducing accuracy or cause a complete loss of localization. Since there is no ground truth to inform the system when the errors was introduced, it is impossible to recover the last valid map state. Therefore, this paper suggests a different approach; instead of updating the map, the environment is analysed to detect changes between what is seen and what is expected. This info is then used to alter the behavior of the localization system - in this case, adaptive MCL (AMCL). To do this, we need a metric that defines how similar (or different) the observed environment is from its representation (the map). However; to the best of our knowledge, no such metric exists. We hypothesize that such a metric can be used to infer the performance of any given localization module that takes the static world assumption, and that it can be used online to adjust its behaviour, thereby reducing the pose error. Furthermore, it can be used to generate a heatmap of an environment, which can then be interpreted by a human operator, in order to identify local regions of high divergence. These are the contributions of this paper:

We propose the Environment Similarity Index (ESI): An index that quantifies how well the currently observed environment matches the map. We also show how the ESI correlates to changes in the environment, both on simulated and real-world data, and how these changes impacts AMCL's performance. In addition, we show how the ESI can be used by a user to easily identify areas in their site that have changed since mapping, and therefore may increase localiztion errors. Finally, we show how the ESI can be used to improve AMCL's performance by using it to dynamically tune AMCL's parameters.

II. RELATED WORK

- Bayes filtering / Probabilistic map update - Scan-BIM divergences - Occupancy grid vs feature maps -

III. METHODOLOGY

This section describes how changes in the environment are defined, and how they are used to calculate the ESI.

A. Changes

The difference that is measured is defined as a *change in how the robot perceives its environment*. The change is measured against the map the robot has of said environment. This is an important definition, with three major aspects. The first is that there can be a lot of changes to an environment, but if the robot does not sense it, then the changes by definition have no impact on the robot's ability to localize itself. The second is that it makes the index agnostic to which localization method is used, and which sensor modalities are available. A difference in observed visual features (I.e. SIFT) can be used just as well as geometric lidar, radar or sonar points, both 2D and 3D. The third is that changes that *could* be sensed but aren't, should not affect the

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index. Changes that happen outside the sensor coverage can naturally not be detected, and so they have no effect on localization. Unseen changes may also change back to the initial state before they are even noticed. The metric should as mentioned apply to any kind of sensors used, but since MiR robots localize via 2D Lidar, this will be the sensor modality used from here on. The reference map could likewise be of any kind (graph based, geometry based, grid based, etc), but for this project, the (occupancy) grid-based map is selected.

Any physical object can result in a change in the environment by being either removed, added, moved, deformed or alter how it interacts with the sensor. However, this can be simplified to just Removed and Added. Any of the last three changes (moved, deformed, different sensor interaction) can be considered as a combination of removing the original item, and adding in the new. The distinction makes no difference from the robots point of view. Finally, an object can be observed to be exactly as expected - this will be denoted as Matched. There are many ways a sensor reading can be categorized as either of these, and the method chosen is by no means the only viable option.

B. The Environment Similarity Index (ESI)

The index is a ratio, based on the current pose estimate from the robots localization module - for this paper, AMCL. On every laser-scan callback, each ray point is evaluated: Is it close to an expected obstacle in the map? If yes, count as a Matched; else, count as an Added. This tells us that an object has been added to the environment that is not on the map. In addition, all laser rays that returned a value are traced in the map from the scanner and outwards to detect obstacles that they *should* have hit but didn't. These objects count towards Removed to indicate that something on the map has been removed from the environment. These classifcations are then used to calculate the ESI score; Figure X provides an overview of the classification.

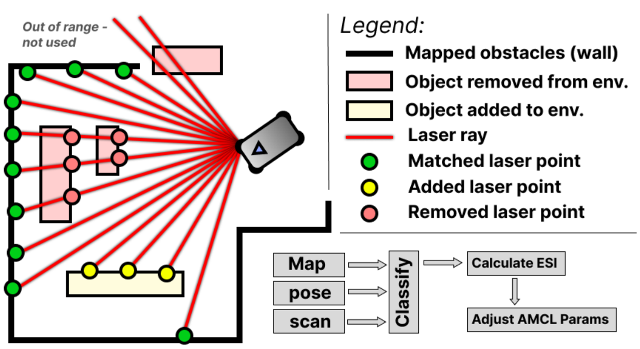

Fig. 1. Example of a figure caption.

C. Formal Definition

Matched: Let S be a 1D vector representing a single scan containing $n_s = |S|$ geometric scan points $s : S = (s_0, s_1, \ldots, s_{n_s-1})$. Only scan points that return a value within min and max range are considered valid. The points are indexed by i where $0 \leq i < n_s$. Let G be a 1D vector representing the discrete geometric reference map, flattened from its original N-dimensional form:

$G = (g_0, g_1, \ldots, g_{n_g-1})$ The occupancy value at index i is $v(g_i)$, which equals 1 if g_i represents an obstacle, and 0 otherwise. The assignment $\lambda(i)$ returns the index in G with the shortest distance to the scan point s_i:

$$\lambda(i) = \underset{0 \leq j < n_g}{\arg\min}\ d(s_i, g_j) \tag{1}$$

The distance $d(p_1, p_2)$ is the Euclidean distance between two points in R^N, where N depends on the map representation, but any distance metric can be used. Let T be a variable distance threshold. It is based on the expected yaw noise γ [rad] and the range of the scan point $r(s_i)$:

$$T(i) = \sin(\gamma) * r(s_i) \tag{2}$$

Define a match as:

$$\chi_M(i) = \begin{cases} 1 & \text{if } d(s_i, g_{\lambda(i)}) < T(i), \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

The number of matches for a single scan is then:

$$M = \sum_{i=0}^{n_s-1} \chi_M(i) \tag{4}$$

Added: All valid scan points are classified as either Match or Added. Therefore, the number of added is simply the number of valid scan points minus the number of matches:

$$A = n_s - M \tag{5}$$

Removed: Each valid scan point defines the endpoint of a line from the scanner to that point, as seen in the world frame. The laser ray L contains all the discrete map points in G that lie on that line: $L = (l_0, l_1, \ldots, l_{n_r-1})$ where l_0 is closest to the scanner. We define a removed point to be every rising edge (going from free to occupied) that the ray encounters from the scanner and out. Define removed as:

$$\chi_R(j) = \begin{cases} 1 & \text{if } v(l_j) = 1 \text{ and } v(l_{j-1}) = 0 \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

The count of removed items (first encounters) in a scan with k rays is then:

$$R = \sum_k \sum_{j=1}^{n_r^{(k)}-1} \chi_R^{(k)}(j) \tag{7}$$

ESI: The ESI score for a single scan S is then defined as:

$$ESI \in [0, 1] = \frac{M}{M + A + R} \tag{8}$$

and since $n_s = M + A$, the final expression becomes:

$$ESI = \frac{\sum_{i=0}^{n_s-1} \chi_M(i)}{n_s + \sum_k \sum_{j=1}^{n_r^{(k)}-1} \chi_R^{(k)}(j)} \tag{9}$$

If the scan only sees matches, the ratio is $1/1 = 1$, indicating a perfect match with the map. Both added and removed points increases the denominator equally, driving the index down. Note how this reflects reality; there is no upper limit to how much you can change an environment; you can always use a sensor with a greater range and higher resolution and then add more items that are not mapped. The index therefore converges towards zero as more changes are made, but it will only be exactly zero if it sees zero matches. The index can be calculated for any number of measurements per sensor update, and can be used for any given localization module. It is important to note that since the ESI depends on the current pose estimate, it will of course not be a correct representation of the environment if the pose error becomes too great. In order to confirm it's usability, we must show that the ESI accurately reflects the environment while the pose error is in its normal range. Then we need to show that ACML is able to maintain its pose error for no, light and moderate changes in the environment.

IV. Experiments

A. *Real-world experiment*

To demonstrate that the ESI works on a real robot in a real enviroment, and that it can be used to visualize changes in it, we used a MiR100 AMR as the platform. It has two SICK S300 safety rated 2D Lidar sensors that scans in a horizontal plane 20cm above the ground, with a 360 degree field of view and a max range of 30m. The venue is the production hall of a packaging and labeling company. It was chosen because it represents the type of environment where AMCL often struggles; it is constantly developing, and most of the time, the the majority of what the robot is able to see consists of transient objects, as seen in Figure X.

*1) **Setup**:* A small section of the hall was mapped, using the native mapping functionality (based on ROS1 cartographer). Two waypoints, Start and Finish, were placed. The task of the robot was simply to move from Start to Finish. In order avoid disrupting their production, instead of moving the items around, we first recorded the map, and then gradually edited it manually, resulting in similar map-environment disagreements. Some transient objects were removed, and new ones added. The robot was tested in four scenarios: 1) No, 2) light, 3) moderate and 4) severe changes. Each scenario was executed 5 times, and the ESI recorded for all runs. When the robot reached the end, its pose error was obtained from a reference coordinate system on the floor. The ESI scores found during each run were then used to create heatmaps for each scenario, as seen in figure X. The method used is Inverse Distance Weighting (IDW) with a power of 2 and radius of 30m. A Line-of-sight (LoS) check was implemented in order to reflect how readings on either side of an obstacle should not affect each other.

TABLE I

ESI results for the real-world experiment. 1) No changes, 2) Light changes, 3) Moderate changes, 4) Significant changes. Rows marked with * have units [mm] and [°], respectively.

Scenario	**1)**		**2)**		**3)**		**4)**	
*** Pose RMSE**	22.4	2.4	17.5	2.1	21.1	2.4	N/A	N/A
*** Pose Std. dev.**	25.1	2.7	19.5	1.1	20.5	1.0	N/A	N/A
ESI mean	0.94		0.64		0.46		0.22	
ESI std. dev.	0.04		0.12		0.10		0.09	

Fig. 2. Photo of a MiR100 moving through the production hall towards the finish waypoint.

*2) **Results**:* A Levene's test of equality of variances on the position errors (Euclidean distance from goal) for scenarios 1-3 shows no significant differences in variance (p = 0.8153, center=median). A one-way ANOVA test shows no significant differences in mean position error (p = 0.7046). These tests show that, in this experiment, for no, light and moderate changes, AMCL is able to consistently maintain a low pose error. However, when the changes became severe, AMCL failed in every run, and the robot was unable to reach the goal. This corresponded with a low mean ESI score of 0.22 for scenario 4.

The heatmap clearly shows how the ESI reflects the intensity of local changes across the hall in the different scenarios.

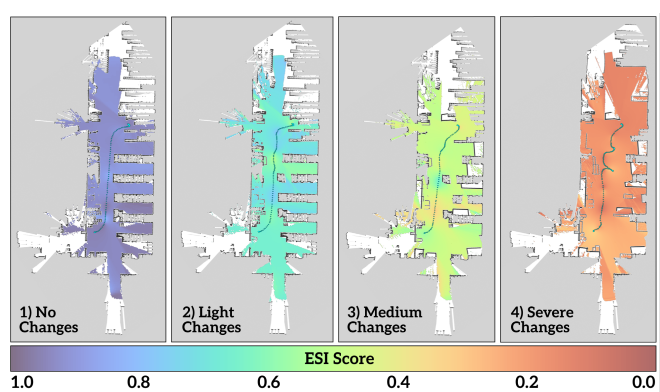

Fig. 3. ESI-based Heatmaps for each of the four scenarios in the same section of the hall.

B. Simulation experiment

A simulated test was performed to demonstrate the following:

- That AMCL provides reliable pose estimates in environments with low to moderate changes.
- How the ESI can be used to quantify changes in the environment, and to what degree it accurately reflects these changes.
- That the ESI can be used as an indicator of when AMCL can no longer be trusted to provide accurate pose estimates due to map-environment differences.
- How the pose estimation error in rapidly changing environments can be reduced by using the ESI to dynamically tune AMCL's parameters.

*1) **Setup**:* The simulator used is Nvidia Isaac Sim v5.0. The robot stack runs on a separate docker container, and communicates with the simulator via ROS2 (humble). The map is 50x50m, and contains 9 waypoints in a grid pattern. The direct pathways between waypoints are considered free space - obstacles are allowed to be placed in the remaining space, as seen in Figure X (red polygons). The experiment will consist of 30 individual runs. In each run, 20 boxes will spawn at random places with random orientations on the obstacle space. An occupancy grid map is then generated and used by the stack, which runs the default AMCL of Nav2 (ref). The robot will move randomly between waypoints, with random odometry noise added. While it moves, every fourth second, a randomly chosen box is moved (teleported) to a new random position in the obstacle space. This continues until all boxes have been moved. Then they are moved back into their original positions one by one, also every fourth second. The test ends when the last box is back in place. This recreates a scenario where the robot moves from a known to an unknown area, and then back again, meaning that in the middle of the test, the map-environment differences are at their highest, as none of the boxes are at their original positions. This is to capture not only when AMCL diverges, but also to see if it is able to recover. The robot has a single 360deg FoW 2D laser scanner with a max range of 30m. As such, it cannot cover the entire map from any one position. The "true" state of the environment is calculated as

$$\frac{U}{U+A+R} \tag{10}$$

where U is the number of boxes left at their original positions, A is the number of boxes added, and R is the number of boxes removed. It is published as `/map_integrity_ratio` (MIRA), and serves as a ground truth to evaluate how closely the ESI reflects the changes. Each run will be performed twice. One with the default AMCL parameters, and one where the ESI is used to dynamically tune the α-parameters of AMCL. These control the noise added to the motion update step. The hypothesis is that when the environment does not match the map, the particle cloud will continue to expand, because none of them have a significantly higher weight than the others, and therefore the filter will not converge. But if some of the particles on the edge of the cloud, where the accumulated random noise is highest, suddenly have just a few beams with a high weight (false positive), the filter will quickly converge to that wrong pose. By using the ESI to limit this expansion, this risk is reduced. Essentially, AMCL now places more trust in the odometry, because the laser data can no longer be trusted. When the environment slowly begins to match the map again, the ESI will rise, returning the filter to its default level of exploration. The two runs are analyzed and compared. Finally, the results are used to develop a method for using the ESI to predict when the pose estimation error will exceed a threshold of 0.5m.

*2) **Results**:* 1. ESI vs map integrity ratio for default AMCL. Evaluation of results in the range of light to severe changes. Figure X (top) shows the individual - and their interpolated mean (black) - ESI scores vs the map integrity ratio, and the absolute position and yaw errors (bottom). In general, the scores follow the MIRA closely. The general pattern and stepwise changes is clearly reflected in the mean, with no notcieable time delay. In the beginning, since the ESI can never be more than 1, the mean is slighly below the true value. It then follows it very closely, and after about 15 seconds, the ESI rises slightly above the MIRA for the rest of the test. Since the robot cannot observe the entire map all the time, this difference is expected. On average, there will be changes outside the range of the laser, which means that the true state of the environment is worse than what the robot can detect. In the final section, the ESI is still above the MIRA, even though the map and environment are now completely different. This is caused by false positive matches that results from the rising pose error, and the fact that the ESI cannot be less than 0. Up until the 50s mark, both position and yaw error is kept low, with mean values of 0.1m and 0.57°, respectively. Yaw does exhibit some spikes to around 10 degrees, but they are quickly corrected. At this point, the ESI is about 0.27 (mir = 0.21). After this, they both start to increase rapidly. This shows that, as long as the map and environment are at least 21-27% similar, the default AMCL is able to provide accurate pose estimates in changing environments. It also shows that, given the robot is well localized, as long as the ESI is kept above 0.27, we can expect it to be an accurate estimate of the map-environment difference. However, there will be scenarios where AMCL has diverged, even with high ESI scores, due to other factors such as map symmetry, lack of features, hallway areas, etc.

Figure caption: Note that 50% of data is used.

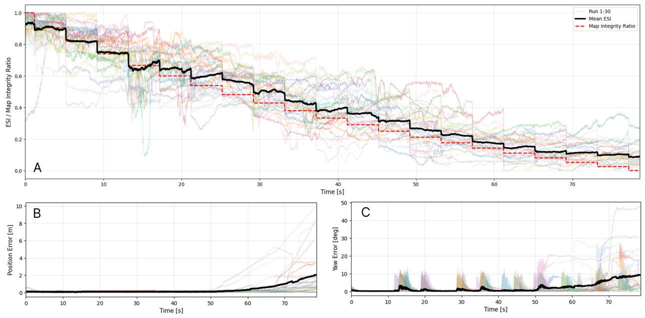

Fig. 4. ESI vs map integrity ratio for default AMCL.

V. Conclusions

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VI. Limitations and Future work

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Appendix

- Raw position data for the real-world experiment -

Acknowledgment

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